Pavan Kumar Etta

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SKILLS

Languages: Python, Go, Java, JavaScript, C, C++, SQL, C#, Kotlin, R, .NET
Backend & Testing: Django, Spring Boot, Node.js, Spark, Test-Driven Development (TDD), Selenium, JUnit
Databases & Tools: DynamoDB, Redis, Prometheus, PostgreSQL, MongoDB, Elasticsearch, Splunk, Datadog, Grafana
Web Technologies: HTML/SCSS, React.js, React Native, Angular.js, Next.js, Vue.js, GraphQL, Flask, Redux
Frameworks: NumPy, Pandas, scikit-learn, PySpark, Hadoop, NLTK, OpenCV, PyTorch, TensorFlow, Agile
Cloud & DevOps: AWS (EC2, Lambda, S3, RDS, API Gateway, EKS, MSK, CloudWatch), Docker, Terraform, Jenkins, Kafka

EXPERIENCE

Graduate Research Assistant — **Mar 2025 – Dec 2025**
University of Massachusetts — Python, PySpark, AWS, Snowflake, Airflow, PyTorch — *North Dartmouth, MA*

- Rearchitected financial-document processing by replacing **Hadoop**-based batch workflows with a serverless, event-driven pipeline using **Python, PySpark, AWS Glue, S3, Athena,** and Step Functions, reducing latency by **50%**.
- Developed and integrated a production document-routing service using **Python, BERT,** and **PyTorch, achieving 93%** accuracy through input validation, error handling, workflow orchestration, and downstream service integration.
- Optimized **Snowflake** schemas and built transformation and scheduling pipelines with **dbt, Airflow, and Python,** reducing financial-report query time by **60%** across data-access workflows.

Software Engineer — **Sep 2022 – Jul 2024**
Rhenix Lifesciences LLP — Java, Spring Boot, Go, AWS, Kafka, Kubernetes, PostgreSQL — *Telangana, India*

- Owned delivery of reusable integrations for five wearable providers **Fitbit, Garmin, WHOOP, Apple Health, and Google Fit** cutting onboarding time by **50%** using **Java, Spring Boot**, **OAuth 2.0, Kafka, PostgreSQL, and microservices**.
- Led webhook scaling to sustain **3x** peak traffic without degradation using **EC2 Auto Scaling**, **Application Load Balancer, Docker, Kubernetes, Redis-backed** workload distribution, and isolated **AWS subnets**.
- Architected a unified clinical model for activity, heart rate, sleep, SpO2, and device metadata, reducing downstream transformation code by **60%** using **C#, .NET, Azure Service Bus, and SQL Server** across workflows.
- Drove observability across synchronization workflows, reducing debugging time by **40%** through **CloudWatch metrics**, **structured logs, audit trails, webhook monitoring**, and **SNS alerts** for token failures, delayed jobs, and malformed events.
- Led modernization of healthcare monolith into scalable **Go microservices** using **gRPC, Kubernetes, and CockroachDB**, reducing deployment time by **65%** and improving reliability across high-volume patient data.
- Spearheaded regression and workflow test automation across engineering, QA, and product teams using **Testim** and **Selenium**, increasing testing speed by **80%**, reducing manual effort and latency by **25%**, and accelerating release validation.

Software Engineer Intern — **Dec 2021 – Jul 2022**
CureScience Institute — Android, BLE, REST APIs, AWS IoT, MQTT, Lambda, DynamoDB — *San Diego, CA*

- Engineered a persistent Android BLE scanning service that enabled **24/7** real-time asset tracking while reducing battery consumption by **85%** and routing location events through gateways and **REST APIs to Amazon S3**.
- Implemented real-time device-heartbeat monitoring through **MQTT, AWS IoT Core, Kinesis, CloudWatch,** and **SNS,** reducing gateway-failure detection from **15 minutes to under 1 minute** using rule-based anomaly alerts.
- Designed and deployed a serverless backend using **AWS Lambda, API Gateway, DynamoDB, RDS,** and **EventBridge**, increasing device-telemetry processing capacity by **3x** across ingestion, persistence, scheduled processing, and asset analytics.

PROJECTS

Fault-Tolerant Sharded Key-Value Store — *C++, Raft, SQLite*

- Engineered a fault-tolerant sharded key-value store in **C++** using **Raft** consensus, **SQLite**, leader election, log replication, and replicated state machines to maintain consistency and support recovery from server failures.

SkillBridge — Real-Time Mentor Matching Platform — *Node.js, Next.js, Socket.IO, Docker, PostgreSQL, AWS EC2, RDS, S3*

- Supported **300** concurrent users and delivered **10,000+** real-time messages with **99.8%** success and under **200 ms p95** latency by optimizing **Socket.IO** connections, PostgreSQL queries, and Docker-based deployment on **AWS EC2** and **RDS**.

Distributed API Gateway & Rate Limiting Service — *Python, Go, AWS, Redis, Docker, Kubernetes*

- Designed a cloud-native API gateway with authentication, request routing, rate limiting, retries, circuit breakers, and service-level observability, improving backend reliability and protecting microservices from traffic spikes.

EDUCATION

University of Massachusetts, Dartmouth — **North Dartmouth, USA**
M.S. in Computer Information Science, Data Science, GPA : 4.0 — *Aug 2024 – May 2026*